

Practical 1: Create a list and perform the following methods append(), insert(), extend(), remove(), copy(), sort(), pop().

```
list_1 = list(("apple", "cherry"))
list_1.append("orange")
print(list_1)
list_1.insert(1, "grapes")
print(list_1)
list_1.insert(4, "papaya")
print(list_1)
list_2 = ["kiwi", "guava", "dragonfruit"]
list_2.remove("guava")
print("Removed Item:", list_2)
list_1.extend(list_2)
print("After Extending:", list_1)
mylist = list_2.copy()
print("After copy:", mylist)
list_1.sort()
print("After sorting:", list_1)
print("Length of list:", len(list_1))
print("Remove Last element:", list_1.pop())
print("After Removing Element:", list_1)
even_numbers = [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22]
num_of_evens = even_numbers.count(4)
print(f"{num_of_evens} number of times")
```

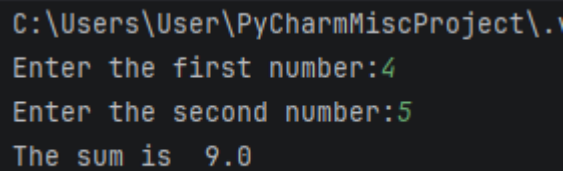
Output:

```
['apple', 'cherry', 'orange']
['apple', 'grapes', 'cherry', 'orange']
['apple', 'grapes', 'cherry', 'orange', 'papaya']
Removed Item: ['kiwi', 'dragonfruit']
After Extending: ['apple', 'grapes', 'cherry', 'orange', 'papaya', 'kiwi', 'dragonfruit']
After copy: ['kiwi', 'dragonfruit']
After sorting: ['apple', 'cherry', 'dragonfruit', 'grapes', 'kiwi', 'orange', 'papaya']
Length of list: 7
Remove Last element: papaya
After Removing Element: ['apple', 'cherry', 'dragonfruit', 'grapes', 'kiwi', 'orange']
1 number of times
```

Practical 2: Write a python program to add two numbers using Function.

```
def add_numbers(num1, num2):  
    return num1 + num2  
  
num1 = float(input("Enter the first number:"))  
num2 = float(input("Enter the second number:"))  
  
result = add_numbers(num1, num2)  
print("The sum is ", result)
```

Output:



```
C:\Users\User\PyCharmMiscProject\..  
Enter the first number:4  
Enter the second number:5  
The sum is 9.0
```

Practical 3: Write a program to create a menu with the following options i) TO PERFORM ADDITION ii) TO PERFORM SUBTRACTION iii) TO PERFORM MULTIPLICATION iv) TO PERFORM DIVISION Accepts users input and perform the operation accordingly. Use functions with arguments.

```
def add(x, y):
    return x+y

def sub(x, y):
    return x-y

def multiply(x, y):
    return x*y

def division(x, y):
    return x/y

num1 = int(input("Enter a First Number="))
num2 = int(input("Enter a Second Number="))

choice = int(input("1. Addition\n2. Subtraction \n 3. Multiplication \n 4. Division \n 5.
Exit \n Enter a choice which operation want to perform[1/2/3/4/5]="))

if choice == 1:
    print("Result=", add(num1, num2))

elif choice == 2:
    print("Result=", sub(num1, num2))

elif choice == 3:
    print("Result=", multiply(num1, num2))

elif choice == 4:
    print("Result=", division(num1, num2))

else:
    exit
```

Output:

```
C:\Users\User\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\User\PyCharmMiscProject\script.py
Enter a First Number=4
Enter a Second Number=6
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter a choice which operation want to perform[1/2/3/4/5]=2
Result= -2
```

Practical 4: Write a program to double a given number and add two numbers using lambda()?

```
num1 = 10
num2 = 20
sum = lambda a, b : a+b
mult = lambda a : a*2

result = sum(num1, num2)
result1 = mult(num1)
print(f"Sum of {num1} and {num2} is {result}.")
print(f"Double of {num1} is {result1}.")
```

Output:

```
Sum of 10 and 20 is 30.
Double of 10 is 20.
```

Practical 5: Demonstrate a python code to implement abnormal termination?

```
def divide(x, y):  
    print(f"{x}/{y} is {x/y}")
```

```
divide(10, 2)  
divide(10, 0)  
divide(10, 4)
```

Output:

```
10/2 is 5.0  
  
-----  
ZeroDivisionError                                Traceback (most recent call last)  
Cell In[17], line 5  
      1 def divide(x, y):  
      2     print(f"{x}/{y} is {x/y}")  
      3  
      4 divide(10, 2)  
----> 5 divide(10, 0)  
      6 divide(10, 4)  
  
Cell In[17], line 2, in divide(x, y)  
      1 def divide(x, y):  
----> 2     print(f"{x}/{y} is {x/y}")  
  
ZeroDivisionError: division by zero
```

Practical 6: Write a python program to get python version.

```
import sys
print("Python Version")
print(sys.version)
print("Version info")
print(sys.version_info)
```

Output:

```
Python Version
3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)]
Version info
sys.version_info(major=3, minor=14, micro=6, releaselevel='final', serial=0)
```

Practical 7: Write a python program to print date, time for today and now.

```
import datetime
current_time = datetime.datetime.now()
print("Current Time is", current_time)

print("The attributes of now() are ")
print("Year:", current_time.year)
print("Month:", current_time.month)
print("Day:", current_time.day)
print("Hour:", current_time.hour)
print("Minute:", current_time.minute)
print("Second:", current_time.second)
print("Microsecond:", current_time.microsecond)
```

Output:

```
Current Time is 2026-07-23 14:49:49.031290
The attributes of now() are
Year: 2026
Month: 7
Day: 23
Hour: 14
Minute: 49
Second: 49
Microsecond: 31290
```

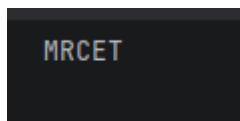
Practical 8: Write a python Program to display welcome to MRCET by using classes and objects.

```
class Display:
    def __init__(self, name):
        self.name=name

    def show(self):
        print(self.name)

d=Display("MRCET")
d.show()
```

Output:

A screenshot of a terminal window with a dark background. The text "MRCET" is displayed in a light blue or cyan monospaced font, centered horizontally.